MIL-Q-10194D

8 September 1969

SUPERSEDING
MIL-Q-10194C
7 February 1963

MILITARY SPECIFICATION

Q-METER TS-617( )/U

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers portable test equipment designated as Q-Meter TS-617( )/U for measuring the Q, or the ratio of reactance to resistance of a coil, capacitor, or other circuit element having two accessible terminals. The Q-Meter consists essentially of an eight-range radio frequency oscillator, a Q measuring circuit consisting of a main and vernier tuning capacitor, a vacuum tube voltmeter, and a means for introducing a known amount of the oscillator voltage in the measuring circuit, all mounted in a suitable cabinet.

2. APPLICABLE DOCUMENTS

2.1 Documents. The following documents of the issue in effect at date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636	Box; Fiberboard
PPP-F-320	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-P-291	Paperboard, Wrapping, Cushioning.
PPP-T-45	Tape; Gummed, Paper, Reinforced and Plain, for Sealing and Securing.
PPP-T-97	Tape; Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-E-75	Electron Tubes, Packaging, Packing and Container Marking for General Specification for
MIL-P-116	Preservation, Methods of
MIL-B-117	Bags, Interior Packaging

FSC 6625

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MIL-P-11268	Parts, Materials and Processes Used in Electronic Communications Equipment.
MIL-M-13231	Marking of Electronic Items.
MIL-F-14072	Finishes for Ground Signal Equipment

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Palletized Unit Loads; 40 in. x 48 in. 4-Way Partial and 4-Way Pallets.
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment.
MIL-STD-252	Wired Equipment Classification of Visual and Mechanical Defects.
MIL-STD-781	Reliability Tests, Exponential Distribution.
MIL-STD-810	Environmental Test Methods.

DRAWINGS

ELECTRONICS COMMAND

SC-D-17269	Caution Plates
SC-A-46439	List of Accessories for Package Tester
SC-DL-183470	Q-Meter TS-617( )/U

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Construction. The equipment shall be constructed in accordance with SC-DL-183470 and the requirements specified herein.

3.2 First article samples. The contractor shall furnish three complete first article samples for approval provided the invitation for bid and the contract requires first article samples.

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3.2.1 Fabrication and assembly. First article samples shall be fabricated by the contractor and shall meet all requirements specified for the equipment on contract. The samples shall be assembled using parts, materials, and processes that will be employed in production, and shall be fabricated and assembled in a manner similar to that to be used in production.

3.3 Parts, materials and processes-general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection of requirements therein, shall apply.

3.4 Finish, protective. The equipment shall be finished in accordance with MIL-F-14072 and the applicable drawing. This includes finish of hardware, such as screws, etc., and necessary touch-up after mounting.

3.5 Marking.

3.5.1 Marking requirement. Marking shall conform to MIL-M-13231.

3.5.2 Technical manual caution notice. Each Q-Meter shall bear a caution notice in accordance with drawing SC-D-17269. Such notice shall appear on the panel that mounts the operating controls of the equipment.

3.6 Preconditioning.

3.6.1 Bounce preconditioning. Each equipment shall be subjected to the bounce preconditioning test of 4.6.1 prior to Group A inspection. After bounce preconditioning, the equipment shall not be repaired, cleaned or otherwise changed prior to performing Group A acceptance test.

3.6.2 Electrical preconditioning. (Burn-in and debugging). Each Q-Meter produced shall be placed in its normal operating position at an ambient temperature of 77°F ±5° with 115 Vac 60 Hz voltage applied. After being operated as specified in 4.7.7.1 to 4.7.7.5 for 24 hours, the equipment shall meet the requirements of 3.7.9. In case of parts failure, the equipment shall be repaired, recalibrated and the burn-in debugging test repeated. A failed analysis report shall be submitted on each part failure with contents of report per 5.10.5 of MIL-STD-781.

3.7 Electrical

3.7.1 Oscillator frequency range. The frequency range of the oscillator shall be variable from 50 kilohertz to 75 megahertz in eight ranges, as shown below. An additional range of 1 to 50 kilohertz is made possible by means of an external oscillator, a shielded connecting cable and an internal jack.

50	to	150 kHz	.....	4.5 to 12.0 MHz
150	to	450 kHz	.....	12.0 to 25.0 MHz
450	to	1500 kHz	.....	25.0 to 50.0 MHz
1.5	to	4.5 MHz	.....	50.0 to 75.0 MHz

3.7.2 Accuracy of frequency calibration. The oscillator frequency dial shall be directly calibrated in frequency to an accuracy of ±1 percent over the range from 50 kHz to 50 MHz and ±3 percent from 50 to 75 MHz when tested in accordance with 4.7.1.

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3.7.3 Range of Q measurements. The total range of circuit Q measurements shall be from 20 to 750. A "Q Range" selector switch shall be provided with ranges of 0-100,0-250, 0-500 and 0-750.

3.7.4 Accuracy of Q measurements. The accuracy of the direct reading measurements of circuit Q (for Q voltmeter readings between Q equals 50 and Q equals 250) shall be ±10 percent of all measurements between 50 kHz and 30 MHz. The accuracy of the direct reading measurement of circuit Q for Q readings between 250 and 750 shall be ±15% for frequencies between 50 kHz and 30 MHz when tested in accordance with 4.7.2.

3.7.5 Injection voltage. The oscillator voltage injected into the Q measuring circuit shall be 20 millivolts rms ±2 percent when the injection level as indicated by the meter reading, is set to the red line. The level setting and measurement shall be made after the zero adjustment is completed and the injection voltage level switch depressed, when tested in accordance with 4.7.3.

3.7.6 Tuning capacitor accuracy. The accuracy of the main tuning capacitor shall be ±1 percent or 1 pF, whichever is greater, for 30 to 100 pF and ±1 percent above 100 pF. The accuracy of the vernier tuning capacitor shall be ±0.1 pF. The integrated capacitor accuracy with respect to dial calibration shall be ±1 percent or 1 pF, when tested in accordance with 4.7.4.

3.7.7 Inductors. Fourteen (14) inductors shall be provided to cover the range from 50 kHz to 20 MHz. The frequency indicated by the frequency dial shall be within 1 percent of the true frequency when the inductor is tuned by the variable capacitor of the Q-Meter to capacitance value indicated on the inductor. The values of inductance and the minimum Q shall be as shown in Table I.

Table I. Inductors

Coil Number	Inductance	"Q"
1.....	25 mh	110
2.....	10 mh	140
3.....	5 mh	160
4.....	2.5 mh	170
5.....	1.0 mh	170
6.....	500 uh	170
7.....	250 uh	200
8.....	100 uh	200
9.....	50 uh	200
10.....	25 uh	200
11.....	10 uh	200
12.....	5 uh	200
13.....	2.5 uh	200
14.....	1 uh	180

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3.7.7.1 Inductor accuracy. The inductors shall be adjusted to an accuracy of ±2 percent of their nominal values, when tested in accordance with 4.7.5.

3.7.8 Input power. The Q-Meter shall be constructed to operate at 115 and 230 volts ±10 percent 49 to 450 Hz power supply and conform to the accuracy requirements of this specification. A switch shall be provided for the selection of 115 or 230 volt power supply. A separate on-off line voltage switch shall also be provided. The power supply pilot lamp shall be a MAZDA Type 43, or equal. (see 4.7.6).

3.7.9 Operational. Each Q-Meter shall be checked to determine accuracies specified in 3.7.2 and 3.7.6 are within tolerances specified when tested in accordance with 4.7.7.

3.8 Service conditions. The equipment shall be constructed so that no fixed part shall become loose, cracked, flaked, pitted or blistered, no movable part or control shall be shifted in setting, position or adjustment, and no permanent degradation shall be caused in performance below that specified herein when subjected to the following conditions.

3.8.1 Temperature. The Q-Meter shall be subjected to temperature extremes specified.

(a) Operating. Temperature extreme of 125°F and -4°F and shall meet the electrical requirements of 3.7.2, 3.7.6 and 3.7.7, when tested in accordance with 4.8.1.

(b) Non-operating. Temperature extreme of 160°F and -80°F and shall meet the electrical requirements of 3.7.2, 3.7.6 and 3.7.7, when tested in accordance with 4.8.1.

3.8.2 Elevation. The Q-Meter shall be subjected to the reduced atmospheric pressure specified.

(a) 10,000 feet operating. Electrical readings shall be taken and meet requirements of 3.7.2 and 3.7.6, at times specified when tested in accordance with 4.8.2.

(b) 50,000 feet non-operating. Electrical readings shall be taken and meet requirements of 3.7.2 and 3.7.6, at times specified when tested in accordance with 4.8.2.

3.8.3 Humidity. The Q-Meter shall exhibit no physical damage or electrical degradation when subjected to the humidity test specified, and shall meet the requirements of 3.7.2, 3.7.6 and 3.7.7, when tested in accordance with 4.8.3.

3.8.4 Vibration. The Q-Meter shall show no evidence of breakage or loosening of parts, and shall meet the requirements of 3.7.2, 3.7.6 and 3.7.7, when tested in accordance with 4.8.4.

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3.8.5 Shock, bench-handling. The Q-Meter shall show no evidence of physical damage, except minor surface abrasion and shall meet the requirements of 3.7.2 and 3.7.6, when tested in accordance with 4.8.5.

3.8.6 Fungus. The equipment shall show no evidence of viable fungus when tested in accordance with 4.8.6.

3.8.7 Visual and mechanical. The equipment shall meet the requirements of MIL-STD-252, as specified in 4.8.7.

3.9 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. Individual items shall not be hand picked for fit or performance. Reliance shall not be placed on any unspecified dimensions, rating, characteristics, etc. Where selection of a component having special characteristics is required, such selection shall be made from qualified or standard parts where possible and the specific characteristics of the selected part shall be defined (see 4.9).

3.10 Technical literature, tools, and running spare parts. Technical literature, tools, and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order (see 6.2).

3.11 Workmanship. The equipment shall be manufactured and assembled in accordance with the applicable portions of the following:

(a) In MIL-P-11268:
General requirements for plastic material and parts.
Wiring and cabling including:
Slack
Protection
Clearance
Connections, general
Grounding, general
Flux and cleaning agents for soldering
Process for soldering
Cleaning of equipment
General process for securing of parts
Process of welding
Connectors, general
Mechanical operation of controls
Tropicalization of material

(b) In this specification:

Marking
Finish

4. QUALITY ASSURANCE PROVISIONS.

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) First article inspection (Does not include preparation for delivery) (see 4.3).

(b) Inspection covered by subsidiary documents (see 4.4).

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.10).

4.3 First article inspection. This inspection shall be performed by the contractor and witnessed by a Government representative unless otherwise specified in the contract. It shall consist of Group A, B and C inspections (see Tables II, III and IV respectively) on three (3) samples of the completely assembled Q-Meter TS-617( )/U. The first article samples shall be inspected in this order: Group A and B for all units; and designated units for specific Group C inspections. Designation of units to specific Group C inspections shall be determined by the Government production engineer. After completion of all Group C tests, conforming units must be reinspected and pass all Group A inspections.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where required
Finish	3.4
Marking	3.5

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4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government - at its discretion - may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor. Each unit, which will be subjected to Group A inspection, shall be preconditioned after final assembly.

4.5.1 Group A inspection. This inspection, including sampling, shall conform to Table II and the ordinary inspection procedures of MIL-STD-105.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in any order which is satisfactory to the Government.

Table II. Group A inspection

Inspection	Req. Para.	Insp. Para.	AQL	
			Major	Minor
Visual and mechanical	3.1, 3.8.7	4.8.7	2.5 dphu*	10.0 dphu*
Electrical				
Oscillator Frequency Range	3.7.1	4.7.1	2.5 dphu* for the group	**
Accuracy of Frequency Calibration	3.7.2	4.7.1		
Range of "Q" Measurements	3.7.3	4.7.2		
Accuracy of "Q" Measurements	3.7.4	4.7.2		
Injection Voltage Calibration	3.7.5	4.7.3		
Main Tuning Capacitor Accuracy	3.7.6	4.7.4		
Inductor Accuracy	3.7.7	4.7.5		
Input Power	3.7.8	4.7.6		
Operational	3.7.9	4.7.7	1.0 dphu	**

* Defects per hundred units.
** All electrical defects are considered major.

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4.5.2 Group B inspection. This inspection, including sampling, shall conform to Table II and to the special procedures for small sample inspection of MIL-STD-105. Group B inspection shall be performed on inspection lots that have passed Group A inspection, and on samples selected from units that have been subjected to and meet the Group A inspection.

4.5.2.1 Order of inspection within Group B. Group B inspection shall be performed in any order which is satisfactory to the Government.

Table III. Group B inspection

Inspection	Requirement paragraph	Test paragraph	AQL
Interchangeability	3.9	4.9	6.5%

4.5.3 Group C inspection. This inspection shall consist of the inspection specified in Table IV and shall be performed on sample units that have been subjected to and met Group A and B inspection. Sample units shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for inspection. Three (3) samples shall be selected at random from every 100 units and fraction thereof produced. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot.

4.5.3.1.1 Contracts for less than 100 units shall have Group C on first production lot as specified in 4.5.3.1.

4.5.3.2 Evaluation of samples submitted to individual Group C inspection. After referenced samples in 4.5.3.1 have been subjected to an individual Group C inspection test and prior to subjection to the next Group C test (unless otherwise specified in the individual test) the sample shall be subjected to and meet electrical tests listed in Table II.

4.5.3.3. Noncompliance. The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, 3 additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Group A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective. (see 6.4).

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4.5.3.4 Order of inspection within Group C. Group C inspection shall be performed in any order which is satisfactory to the Government.

Table IV. Group C inspection

Inspection	Unit number	Req. Para.	Insp. Para.
Temperature	1	3.8.1	4.8.1
Elevation	1	3.8.2	4.8.2
Humidity	1	3.8.3	4.8.3
Vibration	2	3.8.4	4.8.4
Shock, bench handling	2	3.8.5	4.8.5
Fungus	3	3.8.6	4.8.6

4.5.4 Reinspection of conforming Group B and C sample units. Unless otherwise specified, sample units which have been subjected to and passed both Group B and Group C inspection, may be accepted on contract or order, provided all damage is repaired and the sample units are subjected to and pass Group A inspection.

4.5.5 Disposition of Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection, may be delivered on the contract or order, provided units meet requirements of 4.5.4.

4.6 Preconditioning.

4.6.1 Bounce preconditioning. The Q-Meter shall be placed in its normal operating position on the table of a package tester of suitable capacity as made by the L.A.B. Corporation, Skaneateles, New York, with accessories as per Model 1000SC and SC-A-46439, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the equipment under test to a distance of 3 inches as the equipment bounces. The equipment shall be subjected to this preconditioning for 1 minute and shall meet the requirements of 3.6.1.

4.6.2 Electrical preconditioning. (Burn-in and debugging). Each Q-Meter shall be subjected to the test specified in 4.7.7.1 to 4.7.7.5 and shall meet the requirements of 3.6.2.

4.7 Electrical tests.

4.7.1 Oscillator frequency range and calibration.

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4.7.1.1 The output frequencies shall be measured using a Hewlett Packard Type H/P 5247M, or equal, connected across the coil low and ground terminals. The accuracy of the frequency range and calibration shall be ascertained at the end points and four equally distributed intermediate points on each range. Accuracy requirements of 3.7.2 shall be met.

4.7.2 Accuracy and range of Q measurements.

4.7.2.1 Standard coils, Boonton Radio Corporation, Type 513-A and Type 518-A1 through 518-A5, or equal, shall be used to test the center, upper and lower extremes of each range of the Q-Meter up to 30 MHz. Set the range switch to "50-150 kHz", and frequency control dial to the desired frequency setting. Connect the appropriate standard coil between the "Coil Lo" and "Coil Hi" terminals and set the "Level Q" control switch to the "Level" position. Adjust meter to red line with "Set Level" control and with "Level Q" switch to "Q", obtain a peak meter reading by adjusting "Micro Micro Farads - Inductance" fine and coarse controls. Using "Q-Range" control, obtain peak reading. Reading shall meet requirements of 3.7.3 and 3.7.4. Repeat this procedure for the remaining frequency ranges up to and including "12-25 MHz".

4.7.3 Injection voltage calibration.

4.7.3.1 Connect a Boonton Electronics Corporation Model 91CA Meter, or equal, calibrated at the 20 mv point, across the "Coil Lo" and "GND" terminals at frequencies of 50 kHz, 0.5, 5, and 75 MHz. Measurements shall comply with paragraph 3.7.5.

4.7.4 Main tuning capacitor accuracy.

4.7.4.1 Connect the test set to the capacitance measuring equipment as shown in Figure 3.

4.7.4.2 The capacitance measurements will be made at 1 kHz by the substitution method. With the bridge set to substitution, and the cap high lead disconnected from the high substitution input on the bridge, adjust the bridge to read 1010 pF by use of a secondary standard.

4.7.4.3 Set the Q-Meter vernier capacitor at 0.

4.7.4.4 Set the bridge capacitor at the points shown in Part 1 of Table V.

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4.7.4.5 Adjust the Q-Meter main capacitor to obtain a null at each setting of the previous step.

4.7.4.6 Compare the values of step 4.7.4.5, with the values in Table V listed under added capacitance needed for bridge balance. The difference is the main tuning capacitor error and shall be no more than ±1 pF or ±1% whichever is greater. See 3.7.6.

4.7.4.7 Tune the Q-Meter main capacitor to a bridge setting of 970.

4.7.4.8 Set the bridge capacitor at the values shown in part 2 of Table V.

4.7.4.9 Vary the Q-Meter vernier capacitor to obtain a null at each setting of the previous step.

4.7.4.10 Compare the values of step 4.7.4.9 with the values listed in Table V under added capacitance needed for bridge balance. The difference is the vernier error and shall be no more than ±.1 pF. See 3.7.6.

Table V.

	General Radio Capacitance Settings	Added Capacitance Needed for Bridge Balance pF	Tuning Capacitor Allowable Error pF
1	980	30	±1
	978	32	±1
	975	35	±1
	970	40	±1
	950	60	±1
	910	100	±1
	810	200	±2
	710	300	±3
	610	400	±4
	560	450	±4.5
2	970	40	
	969	41	±.1
	968	42	±.1
	967	43	±.1
	971	39	±.1
	972	38	±.1
	973	37	±.1

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4.7.5 Inductor Accuracy.

4.7.5.1 Each coil shall be measured using General Radio Co., Model 1632A Inductance Bridge, for compliance with 3.7.7 and 3.7.7.1.

4.7.6 Input Power.

4.7.6.1 The Q-Meter shall be checked at all combinations of the high, mid and low input voltages and high and low frequencies. The accuracy requirements of 3.7.2 (Accuracy of Frequency Calibration) and 3.7.4 (Accuracy of Q Measurements), shall be met when tested in accordance with 4.7.1 and 4.7.2. The test shall consist of one measurement at each of the combinations required above.

4.7.7 Operational Test.

4.7.7.1 Select one of the 14 inductors furnished and connect it between the "Coil Lo" and "Coil Hi" terminals.

4.7.7.2 Choose the frequency opposite the estimated inductance range as given on the INDUCTANCE-FREQUENCY plate mounted below MICRO MICRO FARADS dial.

4.7.7.3 Adjust the FREQUENCY coarse and fine controls to the frequency determined in 4.7.7.2 above.

4.7.7.4 Throw the LEVEL-Q switch to the Q position.

4.7.7.5 Adjust the MICRO MICRO FARADS INDUCTANCE coarse and fine controls for a peak reading of the meter.

4.7.7.6 Following 4.7.7.5 the requirements of 3.7.9 shall be met. At least one test point from each of the 4.7.1 and 4.7.4 procedures shall be selected to determine proper operation.

4.8 Service conditions.

4.8.1 Temperature test. The following tests shall be conducted with the test set in a temperature altitude chamber in the order specified. Suitable provisions shall be made to pass the required electrical connections through the walls of the chamber. To check temperature stabilization, place a thermocouple in good thermal contact on the largest internal mass centrally located or in any other location deemed necessary by the Government inspector. Temperature stabilization is defined as being accomplished when the temperature of the equipment, verified by three thermocouple readings, does not change by more than $1^{o}F$ in any one half hour period.

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4.8.1.1 With the test set stabilized at an ambient temperature of 25°C. (77°F.), the test set shall be operated for a 20 minute warmup period, to determine whether the performance requirements of the specification are maintained.

4.8.1.2 With the test set not operating, the ambient temperature shall be raised to 71°C. (159.8°F.) and sufficient time for temperature stabilization shall be allowed.

4.8.1.3 The ambient temperature shall be reduced to 55°C. (131°F.) and sufficient time for temperature stabilization shall be allowed. It shall be determined whether the performance requirements of 3.8.1 are met.

4.8.1.4 With the test set not operating, the ambient temperature shall be lowered to -62.2°C. (-80°F.) and sufficient time for temperature stabilization shall be allowed.

4.8.1.5 The ambient temperature shall be raised to 0°C. (32°F.) and sufficient time for temperature stabilization shall be allowed.

4.8.1.6 It shall be determined whether the performance requirements of 3.8.1 are met and all controls and switches can be readily manipulated.

4.8.2 Elevation test. The equipment, placed in its normal operating position in an altitude chamber, shall be tested as follows:

4.8.2.1 With the equipment placed in the chamber, operating and stabilized at room temperature, take the following measurements:

Paragraph	Measurement
3.7.2	Oscillator frequency range and calibration

(a) Reduce the barometric pressure to 20.6 ±0.1 inches of mercury (simulated 10,000 feet above sea level) and, after temperature and pressure stabilization, repeat the measurements specified in 3.7.2. The equipment shall meet specified performance for those measurements.

(b) Remove the power from the equipment and further reduce the barometric pressure to 3.4 ±0.1 inches of mercury (simulated 50,000 feet above sea level). Maintain this condition for 1 hour.

(c) Return the chamber to ambient pressure. Apply power to the equipment and, after stabilization, repeat the measurements specified in 3.7.2. The equipment shall meet specified performance for these measurements.

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4.8.3 Humidity. The Q-Meter in its normal operating position shall be tested as follows and shall meet the requirements of 3.8.3.

4.8.3.1 Apparatus. The chamber and accessories shall be constructed and arranged in such a manner as to avoid condensate dripping on the equipment under test. The chamber shall be vented to the atmosphere to prevent the buildup of total pressure. Relative humidity shall be determined from the dry bulb - wet bulb thermometer comparison method of an equivalent method approved by the procuring agency. When readout charts are used, they shall be capable of being read with a resolution within 1°F. When the wet bulb control method is used, the wet bulb and tank shall be cleaned and a new wick installed at least every 30 days. The air velocity flowing across the wet bulb shall be not less than 900 feet per minute. Provisions shall be made for controlling the flow of air throughout the internal test chamber area where the velocity of air shall not exceed 150 feet per minute. Steam, or distilled, demineralized, or deionized water having a pH value between 6.0 and 7.2 at 73°F. shall be used to obtain the specified humidity. No rust or corrosive contaminants shall be imposed on the equipment by the test facility.

4.8.3.2 Standard ambient conditions. The standard ambient conditions are:

Temperature	+73°F ±5°F
Relative humidity	50 percent ±5 percent
Atmospheric pressure	28.5 (+2-3) inches of mercury

4.8.3.3 Installation of equipment in test facility. The Q-Meter shall be installed in the test facility in a manner that will simulate service usage making connections and attaching instrumentation as necessary. Plugs, covers, and inspection plates not used in operation, but used in servicing shall remain in place. When mechanical or electrical connections are not used, the connections normally protected in service shall be adequately covered. The test chamber shall be at standard ambient conditions when the Q-Meter is installed. The Q-Meter shall then be operated to determine that no malfunction or damage was caused due to faulty installation or handling.

4.8.3.4 Procedure.

Step 1 - Place the equipment in the test chamber in accordance with 4.8.3.3.

Step 2 - Dry the equipment under test at +130°F ±5°F for 24 hours.

Step 3 - Condition the equipment at 73°F ±5°F and 50 percent ±5 percent relative humidity for 24 hours.

Step 4 - Take initial measurements as specified in 4.8.3.5 and inspect the equipment in accordance with 4.8.3.6. The equipment shall meet the specified performance. Record all data necessary to determine compliance.

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Note: The equipment may be readjusted or realigned as necessary to meet specification requirements. No further realignment or readjustment shall be permitted throughout the test period other than normal front panel controls employed for operation of the equipment. No repair or replacement of parts shall be permitted.

Step 5 - Raise the internal chamber temperature to 86°F (30°C) ±5°F.

Step 6 - Subject the equipment to 5 continuous 48 hour cycles in accordance with MIL-STD-170.

Step 7 - After completion of step 6 cycling, condition the equipment for 24 hours at 73°F ±5°F and 50 percent ±5°percent relative humidity.

Step 8 - Inspect the equipment in accordance with 4.8.3.6 within 1 hour.

4.8.3.5 Performance. Measurements shall be completed as rapidly as practicable after the power is applied to the equipment. Measurements required shall be in accordance with 3.8.3.

4.8.3.6 Visual inspection and failure criteria. The equipment shall be visually inspected and a record made of any damage or deterioration resulting from the test. When practicable, visual inspection shall be performed within the chamber and at test conditions. Deterioration, corrosion, or change in tolerance limits of any internal or external part which could in any manner prevent the equipment under test from meeting operational or maintenance requirements shall provide reason to consider the equipment under test as having failed the test.

4.8.3.7 Failure. If any equipment fails to meet the performance specified in 4.8.3.5 after cycling, it does not pass the test.

4.8.4 Test for internal vibration. Internal vibration of the equipment shall be measured to determine conformance to 3.8.4.

4.8.4.1 Secure the equipment directly to a vibration table that can be controlled within 10 percent of the specified amplitude. Mounting method shall be such that vibration within the equipment can be observed and measured. To facilitate this observation and measurement, sub-assemblies may be tested separately provided they are secured to the table in a manner similar to that used to mount them in the equipment.

4.8.4.2 Vibrate the equipment successively in three mutually perpendicular directions over a frequency range of 10 to 55 cycles per second. The total excursion of the applied vibration shall be 0.030 ±0.006 inches.

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4.8.4.3 In each of the three directions, change the frequency in steps of one cycle per second and maintain each frequency for at least 10 seconds.

4.8.4.4 Measure vibration amplitudes by optical means, or by other means provided that vibration of the part is not affected by the measurement.

4.8.5 Shock, bench-handling. The equipment less the transit case and dust cover shall be positioned as for servicing on a solid two inch fir bench top and shall be subjected to the following:

(a) Tilt up the equipment through an angle of 30 degrees, using one edge of the equipment as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axis, for a total of four drops on horizontal face.

(b) Repeat (a) with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the equipment could be placed practicably during servicing.

(c) The equipment shall meet the requirements of 3.8.5.

4.8.6 Fungus. The equipment in its normal operating position shall be subjected to 28 days of fungus test specified in Procedure I, Method 508 of MIL-STD-810 and shall comply with the requirements of 3.8.6.

4.8.7 Visual and mechanical. The equipment shall be examined for the defects listed in MIL-STD-252 and shall meet the requirements of 3.1 and 3.8.7.

4.9 Interchangeability. The dimensions listed below shall be measured to determine conformance to the physical interchangeability requirement of 3.9. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of case, cover and insertable assemblies, when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as case, covers, and mountings.

(d) Location and spacing of test terminals.

4.10 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging of Q-Meter TS-617( )/U and coil case shall be level A or C as specified (see 6.2(b)).

5.1.1 Level A.

5.1.1.1 Cleaning. Each meter and coil case shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each meter and coil case shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packaging. Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be packaged in accordance with item 1 of table VI.

5.1.1.4.2 Q-Meter TS-617( )/U. Each meter and coil case shall be packaged in accordance with items 2 through 16 of table VI and figures 1 and 2.

5.1.2 Level C. Each meter shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2(b)).

5.2.1 Level A.

5.2.1.1 Q-Meter TS-617( )/U. Each Q-Meter TS-617( )/U with coil case packaged as specified in 5.1, shall be packed together within a close fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant, special requirements use. Place the technical literature packaged as specified in 5.1.1.4.1 in the box directly under the lid of the box. Box closure shall be as specified in the appendix to the box specification. To facilitate palletization, fiberboard boxes shall be uniform in size and contain equal quantities of the packaged items to the greatest extent practicable.

5.2.1.2 Palletized load. A quantity of containers packed as specified in 5.2.1.1 shall be placed on a pallet, load type 1, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant W5s. The load shall be "bonded" to the pallet by strapping.

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BILL OF MATERIAL
TABLE VI

REF No.	ITEM Nomenclature	No. Reqd.	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	F	S
1	Technical Literature	2	1C-1	10-1/4	8								
	Bag	2		11-1/2	9		MIL-B-117	II	C				
2	Q-Meter TS-617( )/U containing	1	III	21-1/2	9-3/4	11-1/2	Bare Wt. 35 lbs.						
	Fuse, Spare	2		No Packaging required. Place fuses in clips provided on inside of meter cover. Wind line cord around clips of meter cover, terminating plug in socket on cover. Close cover and secure with latches.									
3	Pad, Top	1		21-1/2	6-1/4	9/16	PPP-F-320	CF	WR	TW			
4	Cell, Top	1		21-1/2	9-3/4	1-3/4	PPP-F-320	CF	WR	SW	W5C		
5	Filler Cell, Bottom-Front	1		21-1/2	3-1/4	3	PPP-F-320	CF	WR	DW	V15c		
6	Pad, Bottom	2		10	4-1/2	3/8	PPP-F-320	CF	WR	SW	W5c		
7	Pad, Back	1		22-1/8	7-1/4	3/8	PPP-F-320	CF	WR	SW	W5c		
8	Pad, End	1		11	7-1/4	5/8	PPP-F-320	CF	WR	SW	W5c		
9	Pad, End	1		12	10-1/8	9/16	PPP-F-320	CF	WR	TW			
10	Sleeve	1		22-3/8	10-1/2	12	PPP-F-320	CF	WR	SW	W5c		
11	Box, Fiberboard	1		22-5/8	10-3/4	12	PPP-B-636	CF	WR	SW	W5c		
12	Case, Coil containing:	1	III	23-7/8	5-5/8	13-3/4	Bare Wt. 34 lbs.						
	Coil, R.F. Plug-in	14		No packaging required. Mount on coil tray.									
	Leg, Metal	2		No packaging required. Place in space provided in case.									
	Cable CG-1432/U	1		2 ft. No packaging required. Place in space provided in case.									
	Cable Assy, Special Purpose	1		No packaging required. Wind cable around clips within coil case.									

Lev. A - PACKED WEIGHT
Lev. A - PACKED VOLUME

T - Type
C - Class
V - Variety
G - Grade
F - Flute
S - Style

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BILL OF MATERIAL
TABLE VI (Cont'd)

REF No.	ITEM Nomenclature	No. Reqd.	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	F	S
	Tube, Electron, Spare	4		Package individually per applicable provisions of MIL-E-75. Store in coil case.									
	Lamp	1	III										
	Paperboard, Cushioning	1		3	2		PPP-P-291	I					
	Tape	1		2	1		PPP-T-45	III			A		
				Roll lamp in paperboard and secure wrap with tape. Store in coil case.									
	Fuse	5	III										
	Paperboard, Cushioning	1		4	3		PPP-P-291	I					
	Tape	1		3	1		PPP-T-45	III		A			
				Space fuses within the corrugations of the paperboard. Roll to compact package and secure wrap with tape. Store in coil case.									
				Packaged tubes, lamp and fuses, shall be stored in the coil case in a manner which will prevent damage or movement during handling and transit.									
	Pad	1		21-1/4	10-1/4		PPP-F-320	CF	WR	DW	V15c		
				Place pad in case between top of coils and lid. Close cover and secure locks.									
13	Cell, Bottom	1		17-3/8	5-5/8	1-1/8	PPP-F-320	CF	WR	SW	W5c		
14	Cell, Top	1		23-5/8	5	1-1/4	PPP-F-320	CF	WR	SW	W5c		
15	Pad, End	2		14	5-3/4	2-1/4	PPP-F-320	CF	WR	SW	W5c		
16	Box, Fiberboard	1		24-1/8	6	14-1/8	PPP-B-636	CF	WR	SW	W5c		

Lev. A - PACKED WEIGHT
Lev. A - PACKED VOLUME

T - Type V - Variety F - Flute
C - Class G - Grade S - Style

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5.2.1.3 Less than palletized load. When quantities per destination are less than a pallet load, containers specified in 5.2.1.1 shall be reinforced by pressure-senstitive filament tape conforming to PPP-T-97, type IV, as specified in the appendix to the box specification. No further packing shall be required.

5.2.2 Level B. A quantity of Q-Meters with coil case packaged as specified in 5.1 shall be packed as specified in 5.2.1.

5.2.3 Level C. A quantity of Q-Meters with coil case packaged as specified in 5.1, shall be packed as specified in 5.2.1, except that pallets, fiberboard boxes and fiberboard caps shall be of the type, size and kind commonly used for the purpose, and shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation. Packing shall be accomplished in a manner that will afford adequate protection to the package and its contents during shipment from the supply source to the first receiving activity.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Q-Meter TS-617( )/U is an instrument which measures the Q, inductance, capacitance and resonant frequency of tuned circuits. It covers the frequency range of 50 kc to 75 mc in eight ranges.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Levels of preservation, packaging and packing (see Section 5).

(c) If rough handling and functional tests are required.

(d) Quantity of first articles required.

(e) Place of final inspection.

(f) Technical literature required.

(g) Quantity of tools and running spare parts required.

6.3 Size data. Size limitations cited herein are based on the dimensions of the equipment for which these requirements were established. Request to exceed these limitations must be supported by data indicating why limitations cannot be met and recommend a revised limitation.

6.4 Nomenclature and type designation. The contractor shall apply for nomenclature in accordance with the applicable clause in the contract. The parentheses in the type designation will be deleted or replaced by a letter identifying the particular design for example: SG-12F/U or AN/URM-73C.

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Q-METER TS-617( )/U

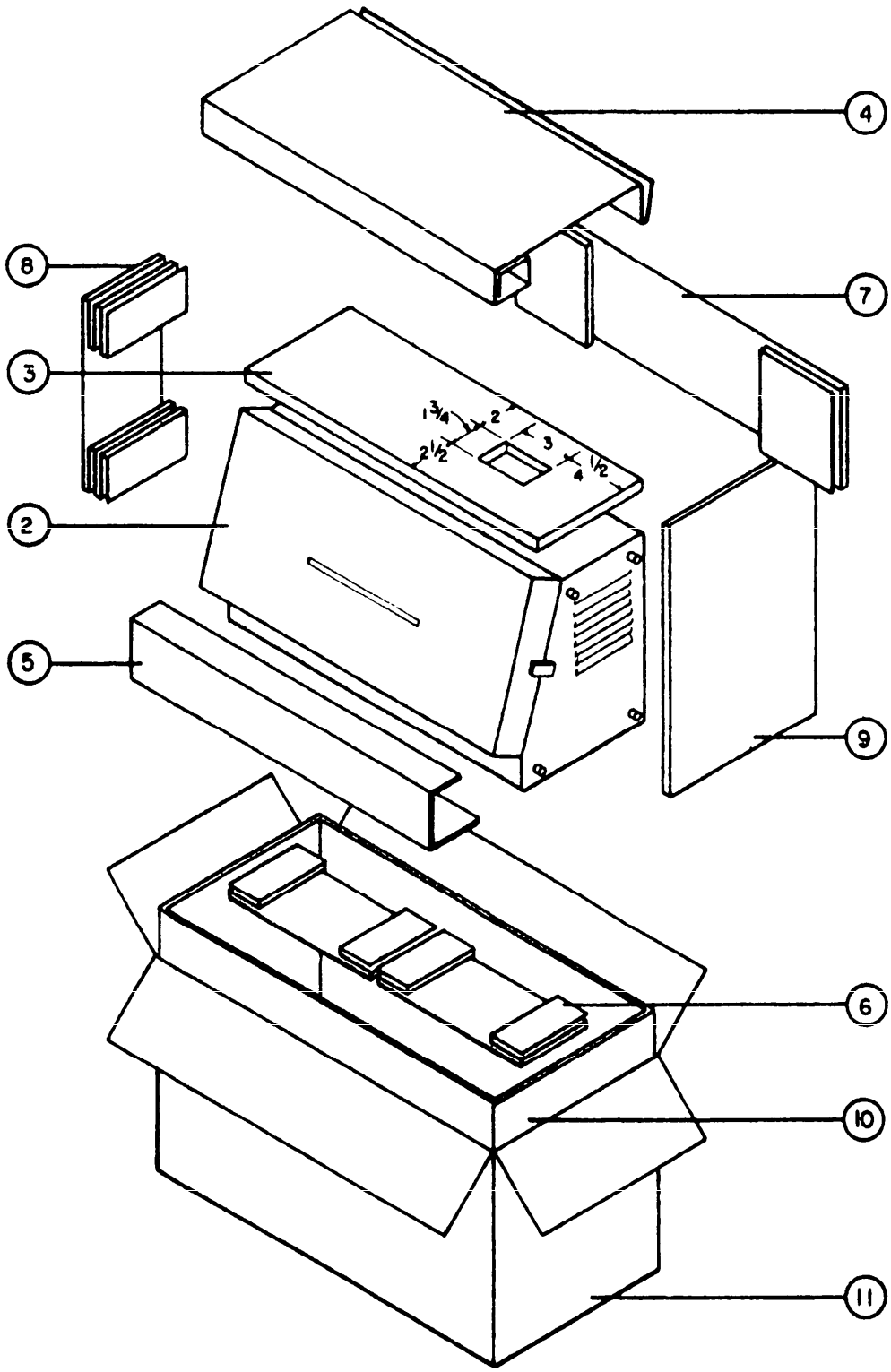


FIGURE 1

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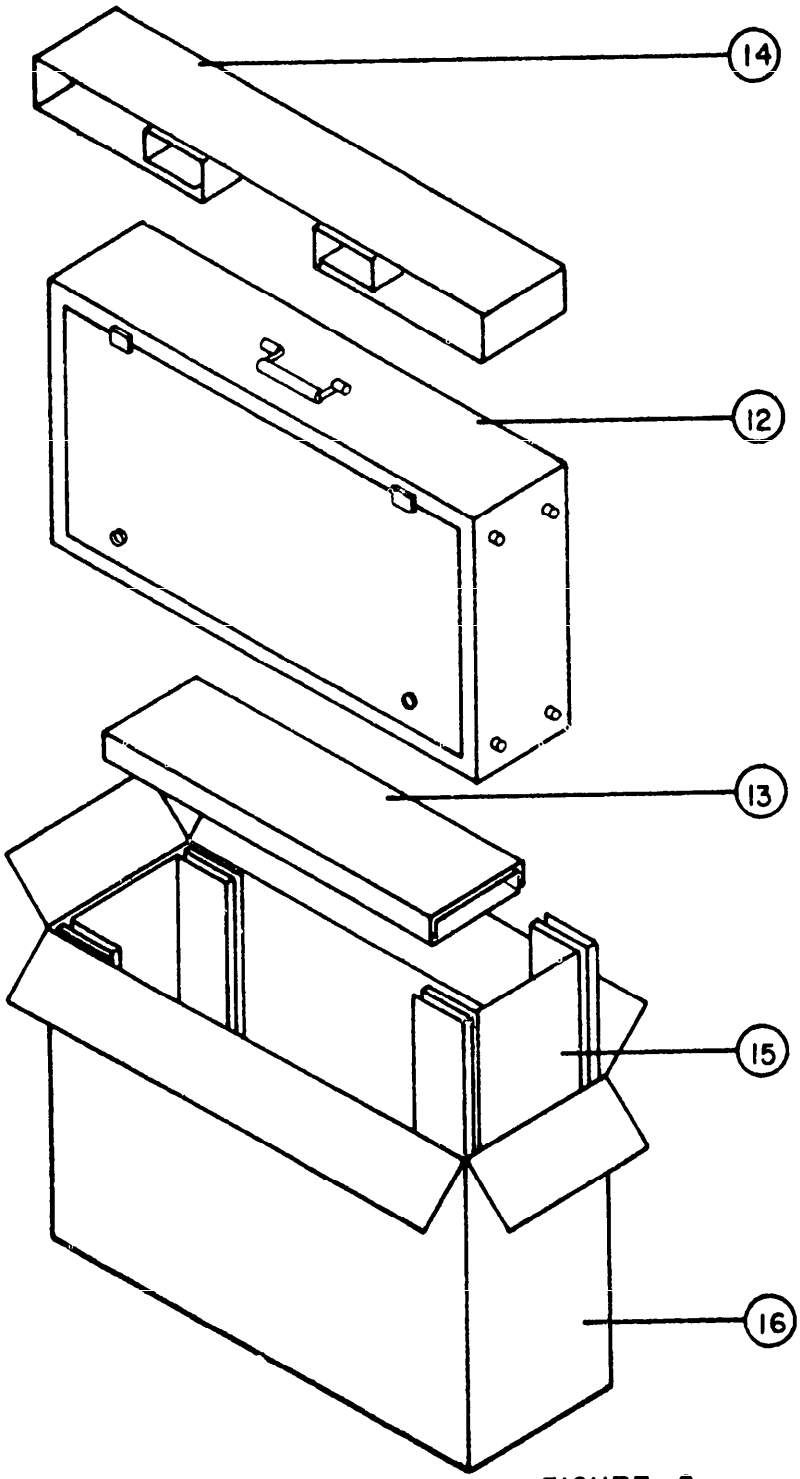


FIGURE 2

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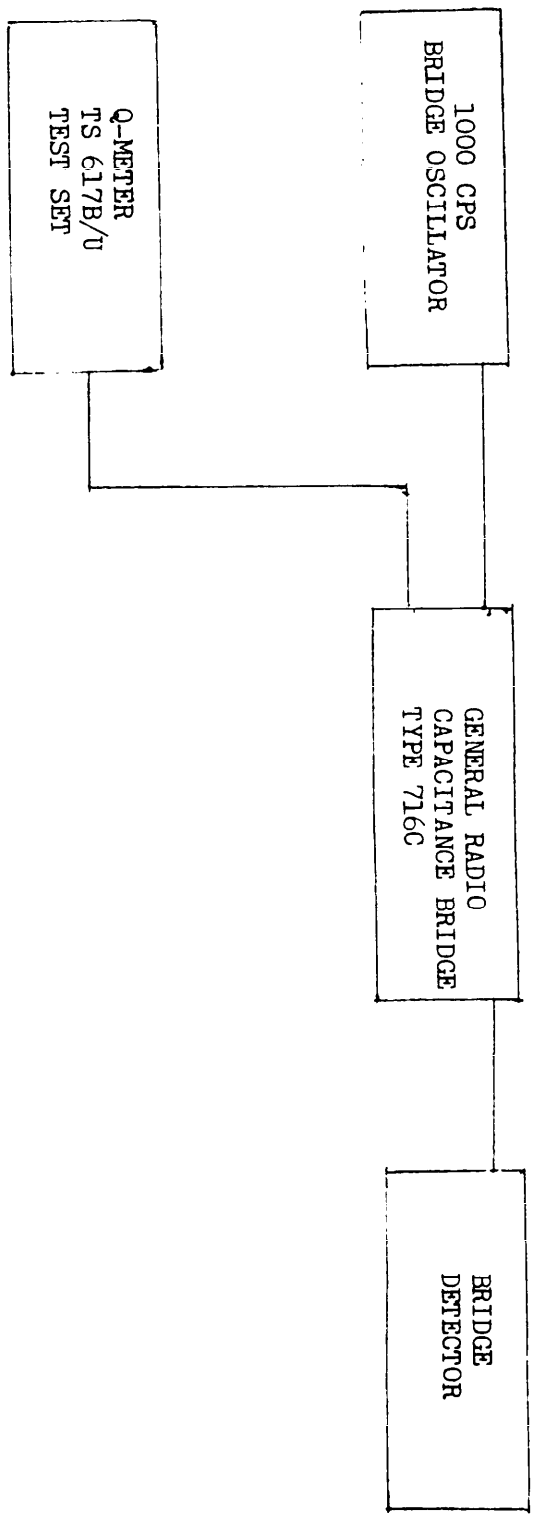


FIGURE #3
TUNING CAPACITOR CALIBRATION TEST CIRCUIT

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6.5 Deviations. Requests for deviations shall be supported by test data proving the ability of the equipment packaged and packed in accordance with the proposed design changes, to withstand the rough handling test requirements of MIL-P-116. Approval will take into consideration the effects on standardization of techniques, availability of packaging and packing materials and economy.

6.6 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) Type A. The total of that inspection set forth in the quality assurance provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by MIL-STD-105.

(b) Type B. That inspection set forth in the quality assurance provisions of this specification or the contract reduced in amount under the reduced inspection provisions of MIL-STD-105.

(c) Type C. A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the quality assurance provisions of this specification. The amount of inspection is less than that provided for in Type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

Custodians:
Army - EL
Navy - EC
Air Force-82

Review activities:
Army - ME, MI, MU, AT, AV, WC
Navy -
Air Force -

User activities:
Army -
Navy -
Air Force -

Preparing Activity
Army - EL

Project Number 6625-0455

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-Q-10194D, Q-METER TS-617( )/U

ORGANIZATION

CITY AND STATE	CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*	DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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